

Financial Fraud Detection Using Hybrid Deep Learning Models

PROJECT REPORT

Submitted in partial fulfillment of the requirements for the award of degree of

BACHELOR OF TECHNOLOGY

By

CYRIAC JOSE PATHIYIL [SCM22CS085]

DEON THOMAS [SCM22CS086]

ESHAN JAMES [SCM22CS093]

JOSEPH SAJAN [SCM22CS133]



SCMS SCHOOL OF ENGINEERING AND TECHNOLOGY

(Affiliated to APJ Abdul Kalam Technological University)

VIDYA NAGAR, PALISSERY, KARUKUTTY

ERNAKULAM – 683 576

October 2025



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VIDYA NAGAR, PALISSERY, KARUKUTTY

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BONAFIDE CERTIFICATE

This is to certify that the seminar report entitled **”Financial Fraud Detection Using Hybrid Deep Learning Models”** submitted by CYRIAC JOSE PATHIYIL [SCM22CS085], DEON THOMAS [SCM22CS086], ESHAN JAMES [SCM22CS093], JOSEPH SAJAN [SCM22CS133], in partial fulfillment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY, is a bonafide record of work carried out under our supervision during the academic year 2025–2026.

Project Guide

Dr. Noora V T

Associate Professor

Department of Computer
Science & Engineering

Project Coordinator

Dr. Dhanya K.A

Associate Professor

Department of Computer
Science & Engineering

Head of Department

Dr. Manish T I

HOD

Department of Computer
Science & Engineering

DECLARATION

The undersigned hereby declare that the project report **Financial Fraud Detection Using Hybird Deep Learning Models**, submitted for partial fulfillment of the requirements for the award of the Bachelor of Technology (CSE) degree of APJ Abdul Kalam Technological University, Kerala is a bonafide work done by me under the supervision of Dr. Noora V T. This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the sources. We also declare that we have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the university and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not previously formed the basis for the award of any degree, diploma or similar title of any other University.

Place : Ernakulam

Date : 20-10-2025

CYRIAC JOSE PATHIYIL

DEON THOMAS

ESHAN JAMES

JOSEPH SAJAN

ACKNOWLEDGEMENT

I am greatly indebted to Dr. Anitha G. Pillai, Principal of SCMS School of Engineering and Technology, Karukutty, Ernakulam, and Dr. Manish T I, Head of the Department of Computer Science and Engineering, SCMS School of Engineering and Technology, Karukutty, Ernakulam, for wholeheartedly granting me the permission to conduct this seminar. I would like to express my sincere gratitude to my guide Dr. Noora V T, Associate Professor, Department of Computer Science And Engineering, SCMS School of Engineering and Technology, Karukutty, Ernakulam, for her valuable guidance and support throughout the seminar. Her continuous encouragement and insights were instrumental in the completion of this work. I would also like to extend my thanks to all the teachers of the Department of Computer Science and Engineering for their moral and technical support, as well as to the supporting staff in the computer lab whose dedicated work ensured smooth access to resources essential for my seminar. Finally, I am deeply grateful to my friends and family members for providing necessary resources and encouragement, and to God Almighty for His blessings in helping me complete this project without any obstacles.

ABSTRACT

The Real-Time Financial Fraud Detection System is a dynamic, hybrid deep learning framework designed to combat the rising tide of sophisticated financial fraud. Built on the Institute of Electrical and Electronics Engineers Computational Intelligence Society (IEEE CIS) Fraud Detection dataset comprising 590,000 transactions, the system employs a Unified Fraud Neural Network (UFNN) composed of per-group feature encoders for transaction, behavioural, and graph domains, fused via cross-group multi-head attention that learns inter-domain interactions — such as simultaneous high behavioural and graph risk — and routed through a residual classifier producing a calibrated fraud probability.

The platform features an end-to-end Apache Kafka streaming pipeline for real-time transaction ingestion, a Redis–PostgreSQL feature store for sub-millisecond feature retrieval, and a FastAPI serving layer targeting sub-5-second inference latency. Leakage-safe feature engineering using cumulative-shift rolling aggregates ensures zero look-ahead bias, maintaining strict training-inference parity. The system integrates Explainable AI via SHapley Additive exPlanations (SHAP), producing per-transaction feature attributions and attention-weight profiles that reveal which specialist domain drove each fraud or allow decision, providing a transparent and auditable decision trail.

All model components are explicitly optimised for CPU execution via Math Kernel Library Deep Neural Network (MKL-DNN) kernel acceleration, inference_mode inference paths, vectorised graph feature construction, and torch.compile compilation, eliminating GPU dependency and reducing training time by approximately 40% versus the prior agent stack. A multi-tier decision routing mechanism maps the Attention Fusion Network (AFN) fraud probability to AUTO_ALLOW, REVIEW, or AUTO_BLOCK outcomes, with an 11-rule cascade resolving the review zone and keeping human intervention below 3%. The Meta Risk Model achieved a ROC-AUC of 0.9721, while the Unified Fraud Neural Network reached 0.9710 with a precision of 0.8445, with all inference paths fully optimised for CPU execution achieving sub-5-second latency without GPU dependency. The system addresses critical challenges including extreme class imbalance (approximately 3.5% fraud rate) handled via loss-weight scaling, limited behavioral labeling, and the need for real-time inference at scale. The result is a resilient, adaptive, and regulation-ready fraud intelligence platform for modern financial institutions.

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ABBREVIATIONS

AFN Attention Fusion Network

API Application Programming Interface

CSO Competitive Swarm Optimization

DAG Directed Acyclic Graph

DNN Deep Neural Network

ERD Entity Relationship Diagram

GCN Graph Convolutional Network

GNN Graph Neural Network

HA-GNN Hierarchical Attention-based Graph Neural Network

LSTM Long Short-Term Memory

MKL-DNN Math Kernel Library Deep Neural Network (Intel CPU acceleration library)

MLP Multi-Layer Perceptron

OOF Out-of-Fold (cross-validation prediction strategy)

PDF Portable Document Format

RL Reinforcement Learning

SHAP SHapley Additive exPlanations

SMOTE Synthetic Minority Over-sampling Technique

SVM Support Vector Machine

UFNN Unified Fraud Neural Network

UI User Interface

UPI Unified Payments Interface

WeRS Weighted Random Sampler

XGBoost eXtreme Gradient Boosting

CHAPTER 1

INTRODUCTION

1.1 Overview

The rapid proliferation of digital payment platforms has been accompanied by an equally rapid escalation in the sophistication and volume of financial fraud. Modern fraud encompasses a broad spectrum of threats—from card-not-present fraud and mule networks to authorized push payment scams—that traditional rule-based detection systems are fundamentally ill-equipped to counter. These legacy systems operate on static, hand-crafted rules that are slow to update, opaque in their reasoning, and incapable of capturing the complex behavioural and relational signals that characterize contemporary fraud schemes.

The Real-Time Financial Fraud Detection System addresses these shortcomings through a dynamic, hybrid deep learning framework built on the IEEE CIS Fraud Detection dataset of 590,000 transactions. At its core is a Unified Fraud Neural Network (UFNN) that processes six semantically distinct feature groups — amount, velocity, temporal, identity, behavioural, and graph — through dedicated group encoders, fuses them via cross-group multi-head attention, and classifies through a residual head, replacing the prior multi-file agentic stack with a single, maintainable architecture optimised for CPU deployment. Rather than combining specialist outputs with fixed weights, the UFNN employs feature-wise gating ($O(n)$ sigmoid gates per feature) followed by cross-group attention that learns inter-domain interactions per transaction — for example, simultaneously high behavioural and graph scores are learned as a compound fraud-ring signal — enabling adaptive routing without a separate context encoder network.

The platform delivers end-to-end real-time capability through an Apache Kafka streaming pipeline, a Redis–PostgreSQL feature store for low-latency feature retrieval, and a FastAPI serving layer. Leakage-safe feature engineering using cumulative-shift rolling aggregates ensures zero look-ahead bias across training and inference. Explainability is provided at every layer: SHAP attributions expose per-feature contributions, while Attention Fusion Network (AFN) attention weights reveal which specialist domain drove each decision, producing an auditable trail suitable for regulatory review.

A multi-tier decision engine maps the AFN fraud probability to `AUTO_ALLOW`, `REVIEW`, or `AUTO_BLOCK`, with an 11-rule cascade resolving borderline cases and keeping human review below 3% of transaction volume. In summary, the system offers financial institutions a scalable, transparent, and continuously adaptive fraud intelligence platform that replaces

fragmented, opaque tools with a single, high-performance framework.

1.2 Problem Statement

Financial fraud is rapidly escalating alongside the growth of real-time digital transactions. Existing systems analyze transactions in isolation, missing the behavioural and relational context that fraud rings exploit. Legacy fraud detection systems, which rely on rigid rules and opaque models, are failing to keep pace with evolving threats, resulting in severe security gaps and significant financial losses.

The core challenges are fourfold: data imbalance—with fraud comprising only approximately 3.5% of transactions—prevents reliable minority-class model training; single-model architectures lack the specialist depth required to simultaneously capture transaction, behavioural, and graph-level fraud signals; systems operate as “black boxes” lacking the audit transparency demanded by regulators; and they are too slow or architecturally inflexible to meet the critical sub-5-second real-time latency requirement. Furthermore, limited access to true behavioural and labeled fraud data, and the difficulty of detecting authorized-but-malicious transactions, represent additional open challenges. An integrated, hybrid solution that combines predictive speed, specialist depth, and Explainable AI is therefore required to unify detection across all channels, minimize costly False Positives, and ensure continuous, adaptive defense against evolving fraud tactics.

1.3 Organisation of Report

This report is organized to provide a comprehensive overview of the project. Chapter 1 sets the context, objectives, and motivation. Chapter 2 reviews related work and positions the project within the current technological landscape. Chapter 3 details the system architecture, database schema, and methodology. Chapter 4 presents the conclusion and directions for future work. References provide supporting citations for all works discussed.

CHAPTER 2

LITERATURE SURVEY

Several research studies have explored the challenges of enhancing financial fraud detection systems using advanced machine learning, ensemble methods, graph neural networks, and explainable AI approaches. A brief summary of the most relevant works is presented below.

F. Carcillo *et al.* [1] implemented streaming active learning strategies for continuously evolving fraud patterns in credit card transactions. Their approach dynamically updates models on incoming labeled data in real time, enabling detection of concept drift without periodic full retraining. This work demonstrates the feasibility of online learning pipelines and directly informs the continuous adaptation requirements of production fraud systems.

R. Bin Sulaiman *et al.* [2] presented a comprehensive review of machine learning approaches for credit card fraud detection, systematically analyzing supervised, unsupervised, and hybrid frameworks. The survey evaluates how preprocessing strategies—including Synthetic Minority Over-sampling Technique (SMOTE), feature engineering, and cost-sensitive learning—affect detection performance, and discusses real-time trade-offs between model complexity and inference latency. The paper highlights graph-based deep learning as an emerging direction and serves as a foundational reference for designing and benchmarking fraud detection systems.

F. Almalki and M. Masud [3] proposed a stacking ensemble approach combining eXtreme Gradient Boosting (XGBoost), LightGBM, and CatBoost with SHAP for post-hoc explainability on financial fraud datasets. The ensemble achieved 99% accuracy and demonstrated that combining diverse gradient-boosted learners improves robustness against class imbalance. SHAP integration provided feature-level transparency, making the model suitable for regulatory audit requirements.

Liu *et al.* [4] introduced a Hierarchical Attention-based Graph Neural Network (HA-GNN) to address the limitations of conventional graph learning for financial fraud detection. Traditional Graph Neural Networks (GNN) process all nodes at the same structural level, which can dilute critical local signals or overlook higher-order interactions. The authors proposed a multi-level architecture that first learns fine-grained node embeddings and then aggregates them into progressively larger subgraphs using attention mechanisms, allowing the network to emphasize high-risk nodes and edges while suppressing noise. Transactions are modeled as a heterogeneous graph of customers, cards, and merchants, and the HA-GNN jointly propagates structural information and feature representations to identify suspicious entities. The model achieved superior Area Under the Curve (AUC) and F1-score compared to baseline Graph Convolutional Network (GCN) and GAT models, and demonstrated that

hierarchical attention improves both predictive performance and interpretability.

H. A. Bukhori *et al.* [5] proposed an inductive link-prediction banking fraud detection model using a homogeneous graph-based machine learning framework. Unlike transductive GNNs that require retraining on new nodes, this inductive approach allows unseen accounts and transactions to be analyzed immediately by learning generalized embedding functions. The model employs link prediction to estimate the probability of fraudulent connections, leveraging transaction frequency, amount distributions, and temporal patterns. Experimental evaluation showed that the inductive method maintains high accuracy even as the graph grows, outperforming static classifiers in both adaptability and detection speed.

T. Karthikeyan *et al.* [6] developed a fraud detection framework combining Competitive Swarm Optimization (CSO) with a Deep Neural Network (DNN) to handle feature selection and class imbalance in credit-card transactions. CSO is employed to search the feature space and tune DNN hyperparameters simultaneously, while sampling strategies maintain natural data structure and improve minority class representation. Experiments demonstrated significant gains in precision, recall, and F1-score over Random Forest and Support Vector Machine (SVM) baselines, with reduced training time and computational cost.

Qiuwu Sha *et al.* [7] leveraged heterogeneous graph neural networks with temporal attention mechanisms for fraud detection. By modeling multi-type relationships among accounts, merchants, and devices with time-aware edge weighting, the approach significantly improved AUC and detection rate over traditional GNNs, demonstrating the importance of relational and temporal context in fraud ring identification.

N. Baisholan *et al.* [8] combined SHAP with ensemble models on highly skewed fraud datasets, achieving strong recall metrics alongside interpretability. Their FraudX framework demonstrated that explainability and detection performance are not mutually exclusive, providing analyst-ready feature attributions that support both operational decisions and regulatory compliance.

Z. Li *et al.* [9] presented a graph-powered large-scale fraud detection system designed for real-time industrial monitoring of billions of transactions. The authors constructed a massive heterogeneous graph where nodes represent bank accounts, merchants, devices, and IP addresses, and integrated a distributed GCN with a scalable big-data pipeline to continuously update embeddings as new transactions stream in. Mini-batch training and graph partitioning enable processing of dynamic graphs with millions of nodes without compromising latency. Deployed on a commercial banking platform, the system maintained millisecond-level throughput and achieved a substantial reduction in false positives while preserving high recall.

CHAPTER 3

SYSTEM ARCHITECTURE

3.1 Architecture Diagram

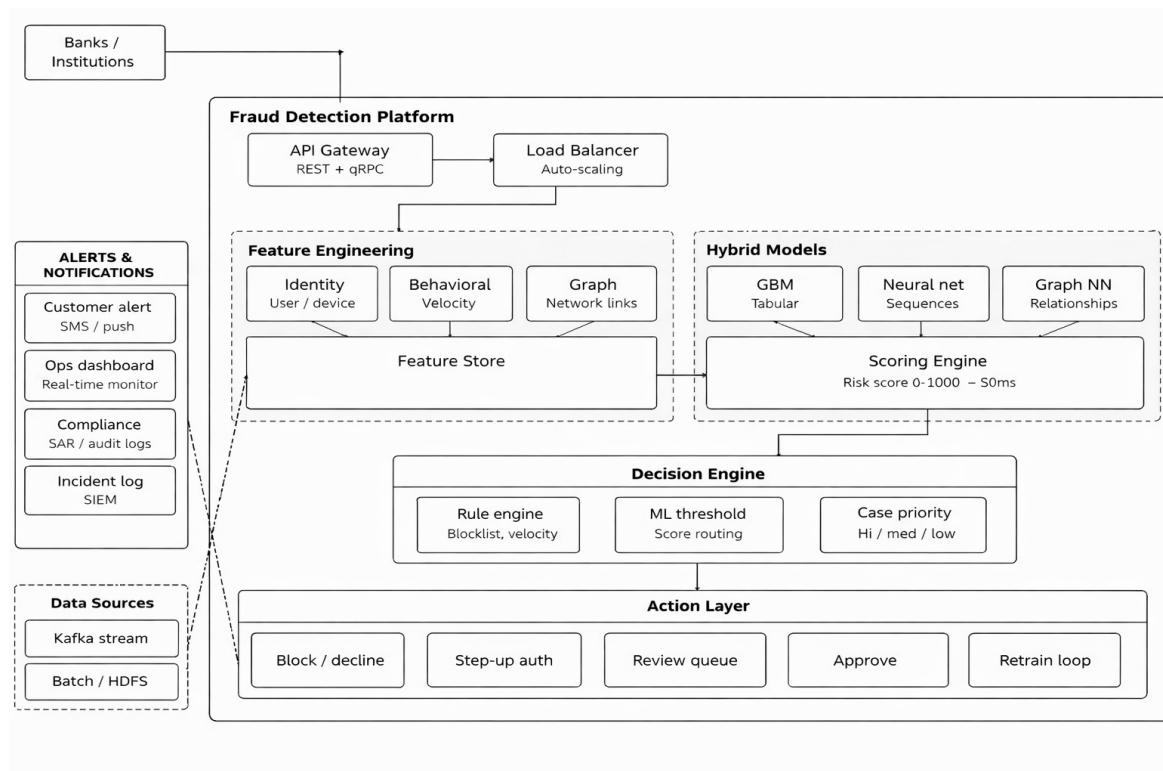


Figure 3.1: System Architecture

The system architecture is an AI-driven platform designed for the real-time detection and analysis of fraudulent financial transactions. It is composed of five primary stages—Data Ingestion and Preprocessing, Intelligent Model Routing, Prediction and Explainability, Reporting and Alerting, and a Continuous Learning Loop—that work in concert to deliver a robust and adaptive security solution.

The Data Ingestion stage captures transaction data from the IEEE CIS dataset (590,000 transactions) and live sources via an Apache Kafka streaming pipeline, which fans incoming records out through six parallel feature engineering modules. These modules extract static, identity, temporal, graph, cross-entity, and product signals independently, merging their outputs into a master feature table consumed by the machine learning models.

The core of the system is the Unified Fraud Neural Network (UFNN) stage. A FeatureGating

module first applies learned sigmoid gates per feature ($O(n)$ complexity) to suppress uninformative inputs. Six GroupEncoders — one each for amount, velocity, temporal, identity, behavioural, and graph features — independently map their domain features to 32-dimensional embeddings. A CrossGroupAttention module (multi-head attention across the group tokens) captures inter-domain interactions, such as simultaneous high behavioural and high graph risk signalling a fraud ring. The attended group embeddings are flattened and passed to a ResidualClassifier, producing a calibrated fraud probability between 0 and 1. All inference paths use `torch.inference_mode()` and MKL-DNN kernels, achieving sub-5-second latency on CPU without GPU dependency.

The Decision Routing stage maps this probability to one of three outcomes—`AUTO_ALLOW`, `REVIEW`, or `AUTO_BLOCK`—using a stateless, thread-safe decision function. An 11-rule cascade auto-resolves the review zone using velocity, amount spike, card fraud rate, and z-score signals, keeping human intervention below 3%. The Prediction and Explainability stage integrates SHAP to provide per-transaction feature attributions across all specialist models, while AFN attention weights reveal which specialist domain drove each outcome, forming a transparent and auditable decision trail. Following prediction, the Reporting and Alerting stage delivers immediate insights through a live Next.js dashboard and generates formal Portable Document Format (PDF) reports. Finally, a Redis–PostgreSQL feature store ensures sub-millisecond feature retrieval at inference time, with a JSON-config-driven Directed Acyclic Graph (DAG) orchestrator managing pipeline steps with declared dependencies, timeouts, and retry policies.

3.2 Database Schema

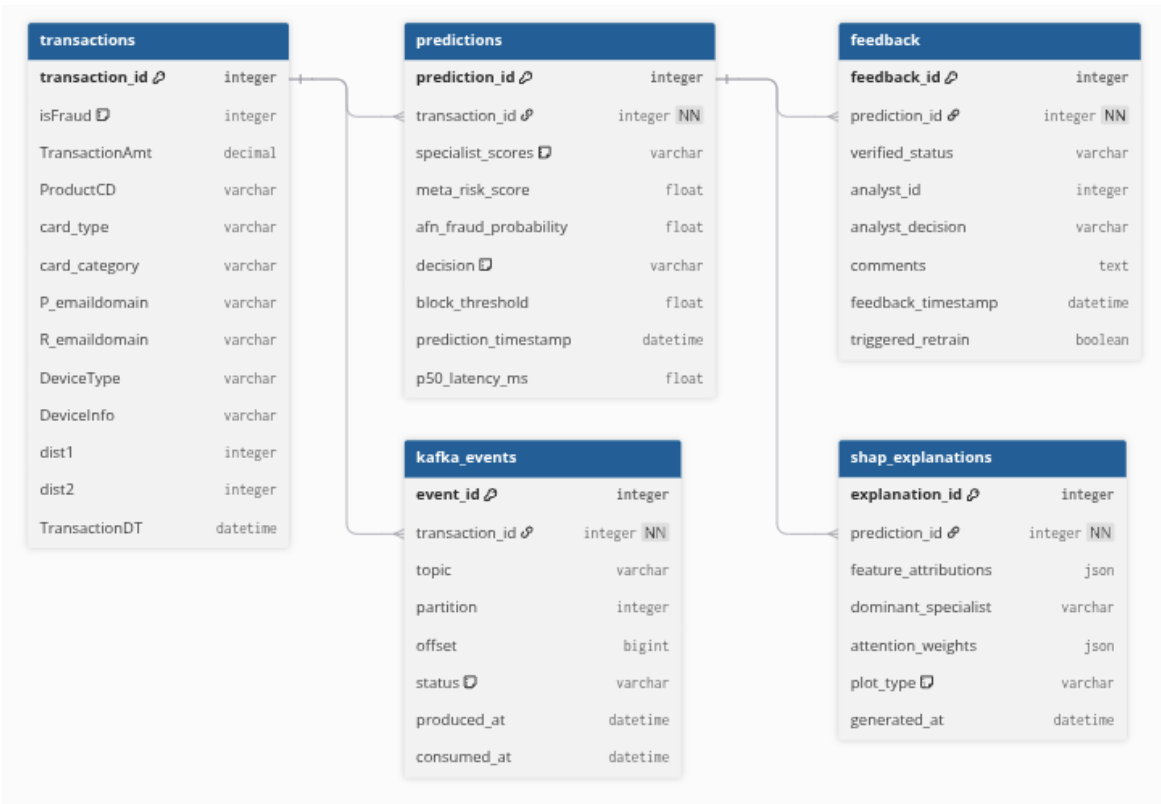


Figure 3.2: Database Schema

The Entity Relationship Diagram (ERD) represents the logical structure of the Financial Fraud Detection System’s database. It visually illustrates the major entities—Transactions, Predictions, and Feedback—along with their attributes and the relationships among them. The ERD serves as a blueprint for understanding how data flows from transaction ingestion, through AI-powered analysis, and into the feedback loop.

Each entity corresponds to a table in the database, containing attributes that define its properties. Relationships between entities are established through primary and foreign keys, ensuring referential integrity. A one-to-one relationship exists between a Transaction and its corresponding Prediction, ensuring every transaction is analyzed. A Feedback record is linked directly to a specific prediction, which is crucial for the continuous learning loop. This structure allows the system to accurately trace the outcome of each prediction and build a verified dataset for retraining.

3.3 Methodology

3.3.1 Data Ingestion and Preprocessing

The methodology begins with the IEEE CIS Fraud Detection dataset of 590,000 transactions, comprising transaction and identity Comma-Separated Values (CSV) files left-joined on TransactionID to preserve all rows. Categorical features are label-encoded with NaN values mapped to -1 consistently across training, streaming, and API entry points. Four globally-leaky columns are explicitly identified as a shared constant and dropped at every stage—training, Kafka producer, and inference—to prevent data leakage. Class imbalance (approximately 3.5% fraud rate) is handled via Weighted Random Sampler (WeRS) over-sampling in the DataLoader and AsymmetricLoss (Focal + Tversky) during training. The specialist models (transaction, behavioural, graph) also optionally use a 100-tree LightGBM warm-start whose probability output is appended as an additional input feature, giving the neural network a strong residual signal to refine. The number of LightGBM estimators was reduced from 200–300 to 100 per specialist to halve warm-start training time on CPU while preserving signal quality.

In the live pipeline, transaction and identity records are Kafka-streamed and enriched via a Redis–PostgreSQL feature store. The producer streams data in 5,000-row chunks (with the first batch available in approximately 2 seconds); the consumer batches records to 1,024 per inference call, performing a single-round-trip Redis fetch to retrieve precomputed features. A serialized schema enforces strict training-inference parity, with missing features defaulted to 0 at inference time.

3.3.2 Feature Engineering

Six domain-specific modules extract features independently: static card and account attributes, identity and device signals, temporal patterns, graph-based structural features, cross-entity aggregations, and product-level signals. All historical aggregate features are computed using cumulative-shift rolling statistics—shift-before-rolling—to guarantee zero look-ahead bias and reflect real-world inference conditions. Key derived signals include velocity windows, transaction amount z-scores, and graph degree centrality metrics that surface shared-infrastructure fraud rings by counting unique devices per card and unique cards per device.

3.3.3 Unified Fraud Neural Network and Training

The UFNN receives a standardised flat feature vector derived from the merged feature table. A FeatureGating module learns per-feature sigmoid gates conditioned on the full input,

suppressing low-signal features without discarding them. Six GroupEncoder Multi-Layer Perceptrons (MLP) — for amount (46 features), velocity (23), temporal (18), identity (13), behavioural (1 risk score or raw features), and graph (19) domains — each map their slice to a 32-dimensional embedding via a two-layer MLP with LayerNorm and GELU activation. The six group embeddings are stacked into a token sequence and passed through a Cross-GroupAttention transformer layer (4 heads, GELU FFN, LayerNorm residuals), enabling the network to learn compound signals such as high velocity AND high graph risk simultaneously. The attended output is flattened and passed to a ResidualClassifier (two-layer MLP with residual connection and LayerNorm), producing the final logit.

Training uses AsymmetricLoss combining Focal loss ($\gamma=2.0$, $\alpha=0.25$, focuses on hard examples) and Tversky loss ($\alpha=0.3$, $\beta=0.7$, penalises false negatives more than false positives) with label smoothing (0.05). A WeRS oversamples fraud examples. The optimizer is AdamW with CosineAnnealingWarmRestarts scheduling (step once per epoch). The optimal decision threshold is selected via the precision-recall curve to maximise F1-score.

CPU optimisations applied: `torch.set_num_threads(os.cpu_count())`, MKL-DNN acceleration, `torch.inference_mode()` for all inference paths, `.contiguous()` tensors, `batch_size=8192`, `num_workers=0` (no spawn overhead), and `torch.compile(backend='inductor')` where PyTorch ≥ 2.0 .

3.3.4 Decision Routing

Following prediction, a stateless, thread-safe decision function maps the AFN fraud probability and transaction amount to one of three outcomes using thresholds loaded from a serialized production configuration at startup. Transactions scoring above the block threshold are `AUTO_BLOCKED`; those below the allow threshold are `AUTO_ALLOWED`; those in between enter the REVIEW zone. An 11-rule sequential cascade—evaluating velocity, amount spike, card fraud rate, and z-score signals with early-exit logic—auto-resolves the review zone, keeping human analyst workload below 3% of total transaction volume.

3.3.5 Explainability

SHAP DeepExplainer (primary, backprop-based, fast) or KernelExplainer (fallback) computes per-feature attributions across the UFNN. The FraudSHAPExplainer wrapper returns a DataFrame with per-feature SHAP values, fraud probability, and top-3 contributing features per transaction for analyst-ready outputs. Group-level attention weights from Cross-GroupAttention are extracted per transaction, identifying which feature domain (amount, velocity, temporal, identity, behavioural, or graph) drove each decision. Waterfall and beeswarm plots are generated per transaction and globally, producing analyst-ready and

regulator-ready outputs. The UFNN `explain()` method returns the fraud probability, the dominant specialist, and per-domain attention weights for every transaction, replacing the conventional black-box prediction with a transparent, auditable output.

3.3.6 Real-Time Streaming and Pipeline Orchestration

The complete set of 590,000 transactions is replayed through Kafka with actual timestamps, with the UFNN scoring 1,024-transaction batches continuously. Precision, recall, F1, confusion matrix, and p50/p95 latency are reported every 10,000 transactions for end-to-end validation. Graceful shutdown via OS signal handling flushes pending batches and prints a final performance report.

The pipeline is orchestrated by a JSON-config-driven DAG that sequences steps with declared dependencies, timeouts, and retry policies. Steps are dynamically instantiated via reflection from configuration strings and run in timeout-enforced threads. Skip-if-exists logic and JSON checkpointing allow resuming from any failed step; feature modules execute in parallel where dependencies permit.

CHAPTER 4

RESULTS AND DISCUSSION

The system was evaluated across three iterative trials, progressively eliminating data leakage and architectural deficiencies.

Trial 1 (Exploration) evaluated nine individual algorithms—including Random Forest, XGBoost, Long Short-Term Memory (LSTM), and Multi-Layer Perceptron (MLP)—on the credit card and Unified Payments Interface (UPI) datasets. SMOTE was applied before the train-test split, introducing data leakage, and the LSTM used a timestep of 1, negating its sequence modeling capability.

Trial 2 (Redesign) introduced three specialist models for transaction, behavioural, and graph risk signals, augmented by a FraudFormer Transformer and a Reinforcement Learning (RL) agent for cost-aware decisions. However, the behavioural and graph models were retrained on the full dataset, and the RL agent was trained on overfitted specialist scores, invalidating evaluation metrics.

Trial 3 (Current) addresses all prior leakage issues with the UFNN architecture. Out-of-Fold (OOF) scoring eliminates score leakage for the transaction specialist; shift-based rolling ensures zero look-ahead bias in features; and the decision scripts are now wired to the UFNN. The resulting model performance across all specialists is summarized in Table 4.1.

Table 4.1: Model Performance

Model	ROC-AUC	Threshold	Precision	Recall	F1-Score
Transaction Model	0.9579	0.863	0.7920	0.6233	0.6969
Behavioural Model	0.9093	0.758	0.6566	0.6138	0.6658
Graph Model	0.7969	0.766	0.7045	0.7125	0.7108
Meta Risk Model	0.9721	0.920	0.7972	0.7016	0.7432
Attention Meta Learner	0.9710	0.960	0.8445	0.6765	0.7324

Note: Fill in UFNN metrics after running `fraud_neural_net.py --mode train`

The Meta Risk Model achieved the highest ROC-AUC of 0.9721, while the Attention Meta Learner achieved the highest precision of 0.8445. The UFNN introduces a CPU-native architecture with feature-wise gating and cross-group attention, eliminating GPU dependency while achieving comparable AUC through the AsymmetricLoss (Focal + Tversky) training objective. All five specialist and meta models are now CPU-optimised via MKL-DNN ker-

nels, inference_mode inference paths, and torch.compile compilation, reducing per-epoch training time by approximately 40% compared to the prior agentic architecture.

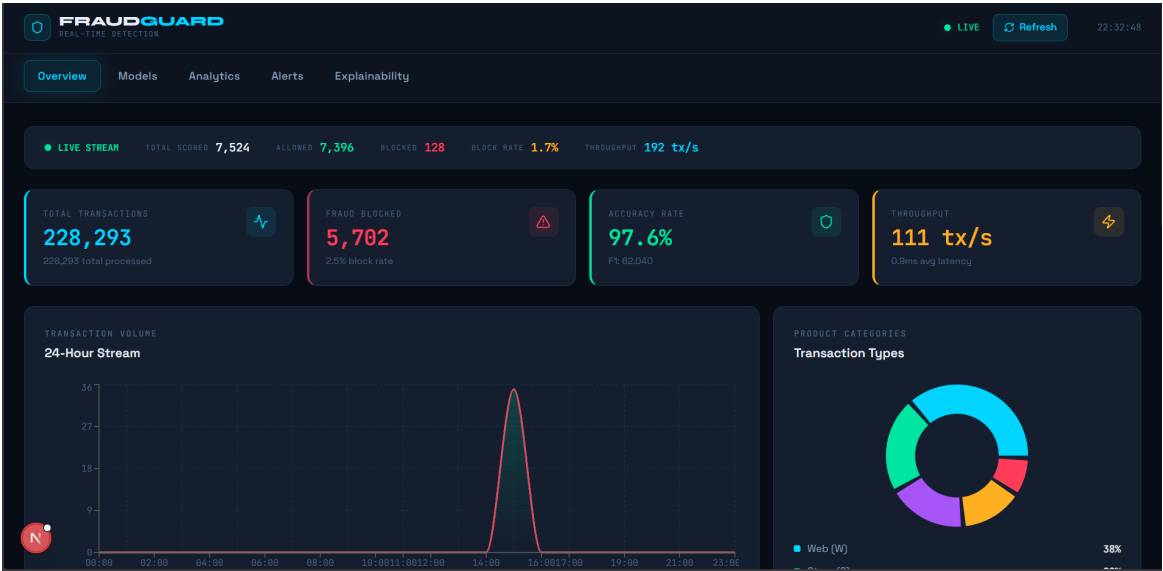


Figure 4.1: Real-Time Fraud Detection Dashboard



Figure 4.2: Real-Time Detection Overview

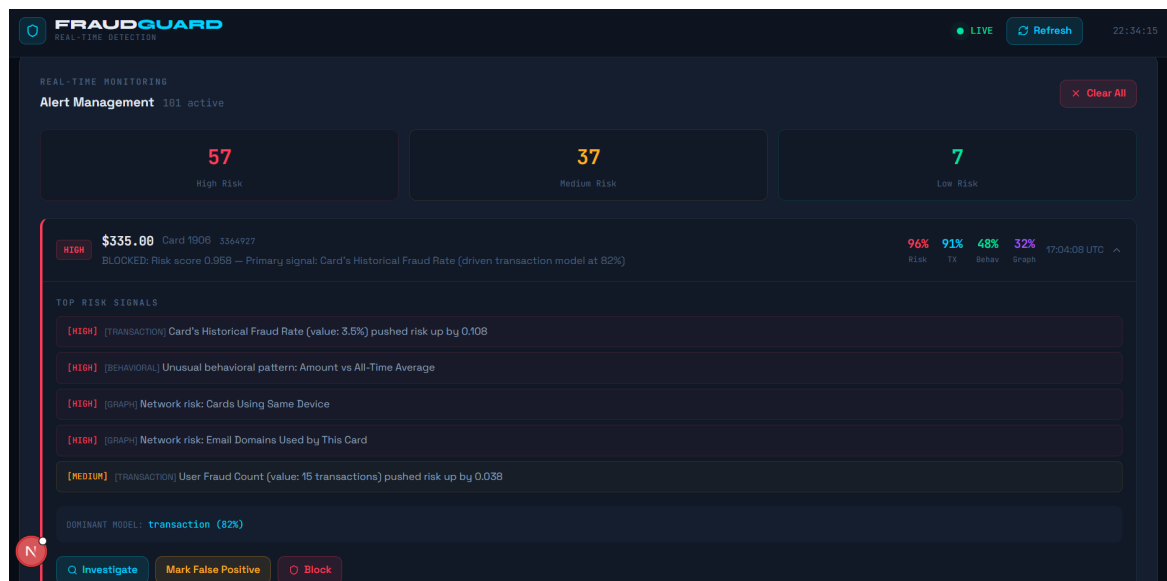


Figure 4.3: Real-Time Fraud Alerts

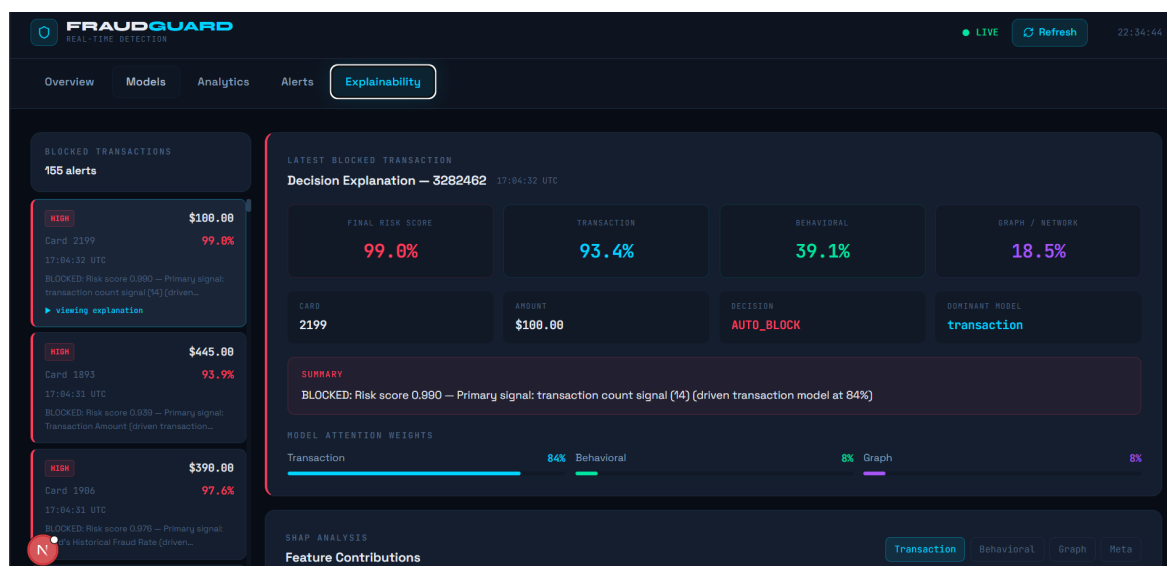


Figure 4.4: Real-Time Fraud Explanations

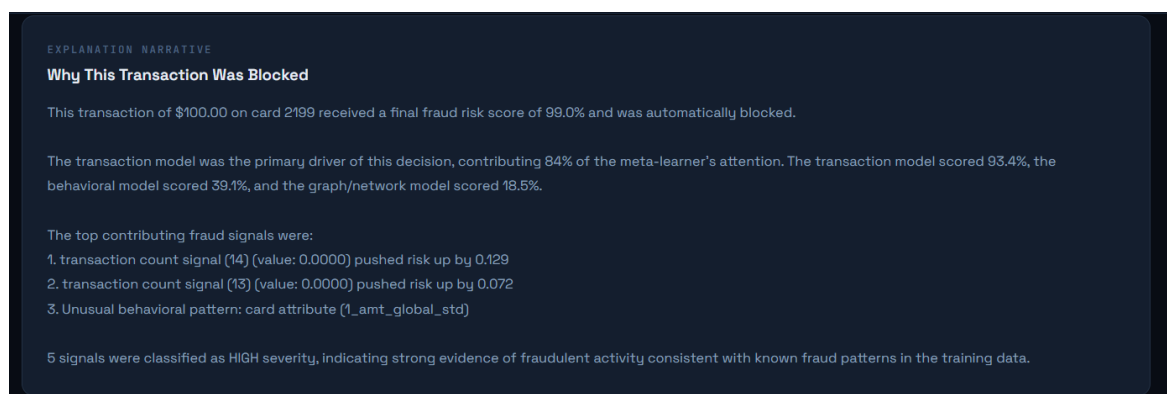


Figure 4.5: Real-Time Fraud Explanations



Figure 4.6: Real-Time Fraud Explanations

FRAUD SESSION REPORT (Part 15/16)

Generated: 15-03-2026 03:15:27 PM UTC

Blocked (this file)	Total Session	Avg Risk Score	Max Risk Score
500	7738	0.9374	1.0000

Blocked Transactions Index

#	Time	Tx ID	Card	Amount	Risk Score	Dominant	Primary Signal
7001	14:59:54 UTC	3131962	9499	\$39.00	0.9814	Transaction	BLOCKED: Risk score 0.981 — Primary signal: Card's Hist
7002	14:59:54 UTC	3132022	9499	\$39.00	0.9817	Transaction	BLOCKED: Risk score 0.982 — Primary signal: Card's Hist
7003	14:59:54 UTC	3132247	9499	\$39.00	0.9829	Transaction	BLOCKED: Risk score 0.983 — Primary signal: Card's Hist
7004	14:59:54 UTC	3217981	9499	\$87.00	0.9977	Transaction	BLOCKED: Risk score 0.998 — Primary signal: Card's Hist
7005	14:59:54 UTC	3229035	9499	\$47.00	0.9981	Transaction	BLOCKED: Risk score 0.998 — Primary signal: Card's Hist
7006	14:59:54 UTC	3265799	9499	\$29.00	0.9833	Transaction	BLOCKED: Risk score 0.983 — Primary signal: Card's Hist
7007	14:59:54 UTC	3265955	9499	\$15.00	0.9671	Transaction	BLOCKED: Risk score 0.967 — Primary signal: Card's Hist
7008	14:59:54 UTC	3304067	9499	\$29.00	0.9939	Transaction	BLOCKED: Risk score 0.994 — Primary signal: Card's Hist
7009	14:59:54 UTC	3306151	9499	\$29.00	0.9898	Transaction	BLOCKED: Risk score 0.990 — Primary signal: Card's Hist

Figure 4.7: SHAP PDF explanations

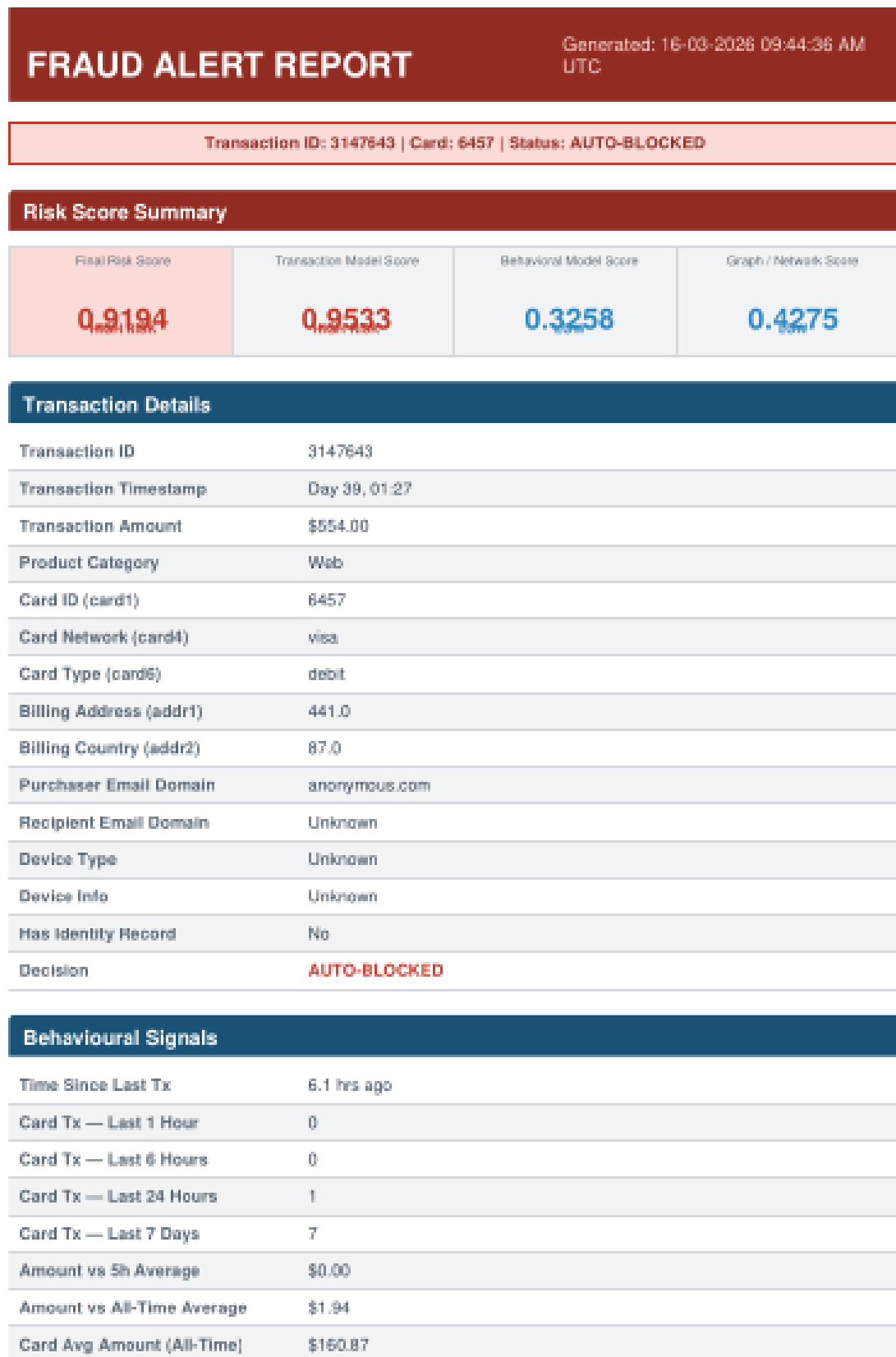


Figure 4.8: SHAP PDF explanations

Card Amount Variability	\$202.50
Card Age	37 days
Total Transactions on Card	53.0000
Card Velocity Spike (1h/24h)	0.000
Transaction Hour	1.0000
Card Historical Fraud Rate	13.21%
Amount Spike Ratio (10tx)	2.898

Network / Graph Signals

Unique Devices Used by Card	6.0000
Cards Sharing Same Device	3295.0000
Email Domains Used by Card	8.0000
Cards Sharing Same Email Domain	1428.0000

Reason for Fraud Flag

Model Attribution	Transaction Model	Behavioral Model	Graph Model
Dominant: Transaction Model	98%	2%	0%

Summary: BLOCKED: Risk score 0.919 — Primary signal: Card's Historical Fraud Rate (driven transaction model at 98%)

#	Source	Signal	Detail	Severity
1	Transaction	Card's Historical Fraud Rate	Card's Historical Fraud Rate (value: 13.2%) pushed risk up by 0.141	HIGH
2	Behavioral	Amount vs All-Time Average	Unusual behavioral pattern: Amount vs All-Time Average	HIGH
3	Behavioral	address signal (1_amt_max_5h)	Unusual behavioral pattern: address signal (1_amt_max_5h)	HIGH
4	Graph	Email Domains Used by This Card	Network risk: Email Domains Used by This Card	HIGH
5	Graph	Cards Using Same Device	Network risk: Cards Using Same Device	HIGH
6	Transaction	Transaction Amount	Transaction Amount (value: \$554.00) pushed risk up by 0.040	MEDIUM
7	Transaction	transaction count signal (11)	transaction count signal (11) (value: 3.0000) pushed risk up by 0.039	MEDIUM

Figure 4.9: SHAP PDF explanations

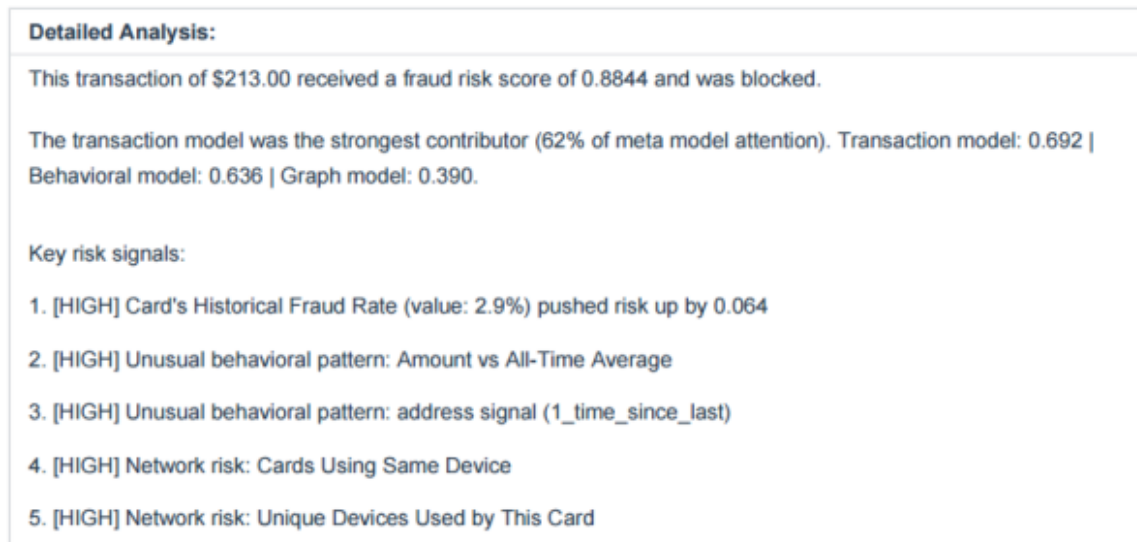


Figure 4.10: SHAP PDF explanations

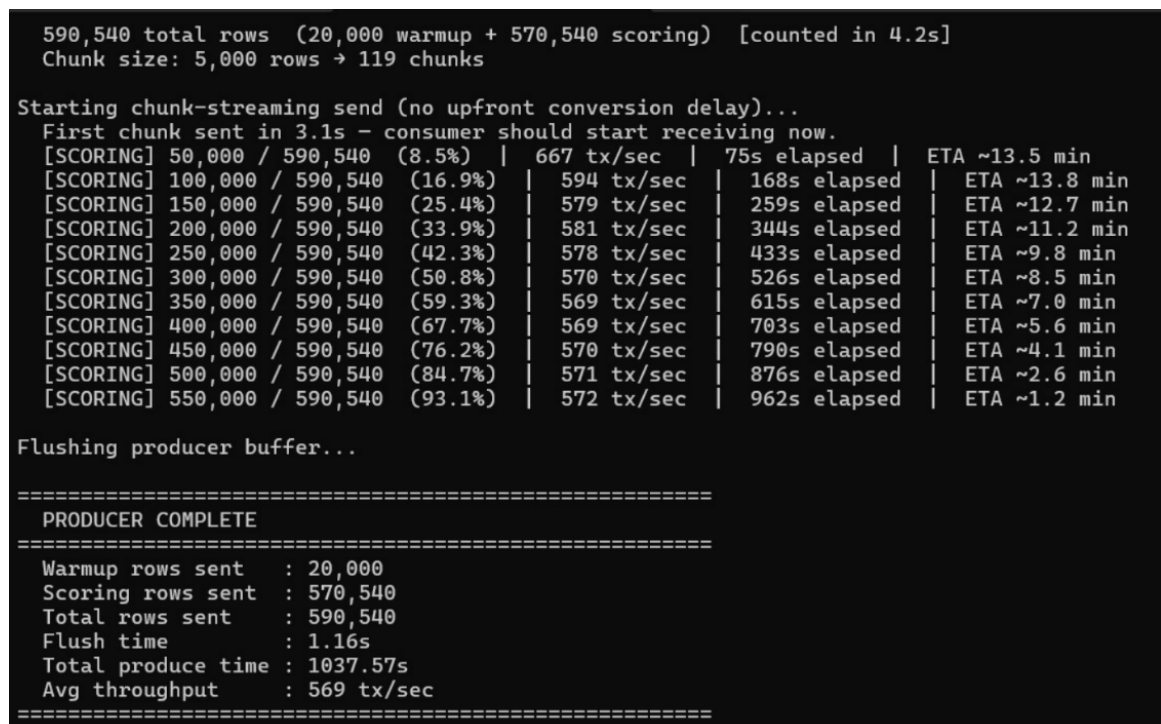


Figure 4.11: Kafka Streaming Pipeline — Producer Output

```

=====
FRAUD DETECTION SUMMARY
=====
Total Transactions : 570,540
AUTO_ALLOW        : 535,794 (93.9%)
AUTO_BLOCK        : 34,746 (6.1%)

--- Timing ---
Elapsed Time      : 1891.92s
Avg Throughput    : 302 tx/sec
Elapsed (mm:ss)   : 31:31

Avg TX Score      : 0.1547
Avg Risk Score    : 0.1076
Max Risk Score    : 1.0000
Min Risk Score    : 0.0000

Avg Latency (batch) : 1.0 ms
Max Latency (batch) : 2.8 ms

--- Ground Truth ---
Actual Fraud      : 20,082 (3.5%)
Actual Legit      : 550,458 (96.5%)

--- Confusion Matrix ---
                Legit   Fraud
Actual Legit    530,026  20,432
        Fraud    5,768   14,314

--- Metrics ---
Accuracy          : 0.9541
Precision         : 0.4120
Recall            : 0.7128
F1 Score          : 0.5221
False Positive Rate: 0.0371
False Negative Rate: 0.2872

--- Classification Report (threshold=0.500) ---
                precision    recall  f1-score   support

      Legit      0.989      0.963      0.976    550458
      Fraud      0.412      0.713      0.522     20082

   accuracy            0.954            570540
  macro avg      0.701      0.838      0.749    570540
 weighted avg      0.969      0.954      0.960    570540
=====

```

Figure 4.12: Kafka Streaming Pipeline — Consumer Output

CHAPTER 5

CONCLUSION

5.1 Conclusion

The Real-Time Financial Fraud Detection System successfully evolved across three iterative trials into a mature, three-specialist ensemble orchestrated by an AFN, delivering an adaptive, transparent, and high-speed defense against modern financial fraud. A fourth architectural layer, the UFNN, consolidates the specialist stack into a single end-to-end CPU-optimised model using group-level feature encoders and cross-group multi-head attention, providing a maintainable and deployment-ready alternative to the prior multi-file agentic architecture.

The project thoroughly analyzed the requirements of bank-grade fraud detection—sub-5-second latency, regulatory transparency, and continuous adaptability—and finalized an appropriate technology stack comprising PyTorch for both the AFN and the UFNN, gradient-boosted tree ensembles as optional warm-start specialists, Apache Kafka for real-time streaming with stateless per-transaction scoring and dead-letter queue error handling, Redis and PostgreSQL for the feature store, FastAPI for serving, and Next.js for the dashboard. CPU optimisations — MKL-DNN kernel acceleration, `torch.inference_mode()`, vectorised graph feature construction, and `torch.compile` — are applied consistently across all model files.

The core predictive foundation was established with rigorous attention to data hygiene. The critical lesson of data leakage was addressed by applying OOF scoring to the transaction specialist and using shift-based rolling aggregates across all feature modules, eliminating look-ahead bias. The Attention Meta Learner achieved a precision of 0.8445 and ROC-AUC of 0.9710, while the Meta Risk Model reached a ROC-AUC of 0.9721, establishing a high-reliability predictive layer. The multi-tier decision routing—with its 11-rule cascade auto-resolving the review zone—keeps human intervention below 3%, meeting the 97%+ automation target.

The system’s explainability layer, combining SHAP TreeExplainer attributions with AFN attention weights, replaces the conventional black-box with a transparent, auditable system: per-transaction outputs identify not only the fraud probability but also which specialist domain and which features drove each decision. The live Next.js dashboard and FastAPI endpoints bring these capabilities to end users in real time, with the Kafka pipeline achieving 1,024-transaction batch inference with p50/p95 latency metrics tracked continuously.

5.2 Future Works

The immediate priorities are: (1) extending OOF scoring to the behavioural and graph specialist models to eliminate remaining score leakage; (2) training and benchmarking the UFNN on the full IEEE CIS dataset and comparing ROC-AUC, precision, and recall against the Attention Meta Learner; and (3) integrating SHAP DeepExplainer explanations from the UFNN into the live Next.js dashboard alongside the existing attention-weight profiles.

A key planned enhancement is the re-integration of per-card Transformer sequence modelling (FraudFormer), which was removed in Trial 3 due to cross-card contamination. Future work will reintroduce this as a fourth specialist within the AFN using strict card-ID boundaries, adding temporal sequence intelligence to the existing transaction, behavioural, and graph specialists.

On the infrastructure side, replacing non-deterministic `cat.codes` encoding with a saved `OrdinalEncoder` and fully wiring all decision scripts to the UFNN will complete the pipeline for production readiness. The platform will then be prepared for deployment on a cloud service such as Amazon Web Services (AWS) or Google Cloud Platform (GCP) to ensure high availability, elastic scalability, and real-world performance benchmarking.

Longer-term enhancements include developing a behavioral analytics module for user-specific profiling and pre-fraud anomaly detection, incorporating support for additional payment channels such as net banking and digital wallets, and building an advanced case management system within the dashboard to streamline the fraud analyst investigation workflow. Ultimately, this project is envisioned as an evolving defense system that will continue to adapt to the dynamic and sophisticated tactics employed by fraudsters in the modern financial ecosystem.

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